TSM 099 Arithmetic Test - Practice 2

Version 202610122020.17 | Name: _____

Complete and show all work on a separate sheet of paper. Record final answers on the answer sheet. Give exact answers and simplify completely.

Answer key follows the answer sheet. Attempt the test before checking it.

1. Evaluate at (a) x = 3 and (b) x = -3. Show substitution.

$(x+3)(2x-2)$

2. Simplify the square root.

$\sqrt{75}$

3. Complete the table by evaluating each side separately. State what the tested values establish; do not solve the equation.

$\frac{x^2-9}{x-3} = \frac{2x^2+11x+12}{x+4}$

x	Left value	Right value	Equal?
7			
-7			
0			

4. Evaluate the principal square root in each part.

(a) $\sqrt{-64}$

(b) $\sqrt{36}$

5. Reduce each fraction to lowest terms.

(a) $\frac{60}{84}$

(b) $\frac{25}{35}$

Student answer sheet

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Question 1

(a) 10 pts

(b) 10 pts

Question 2

Radical 20 pts

Question 3

x = 7 4 pts

x = -7 4 pts

x = 0 4 pts

Conclusion 8 pts

Question 4

(a) 10 pts

(b) 10 pts

Question 5

(a) 10 pts

(b) 10 pts

Answer key

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Question 1

(a)	24	10 pts
(b)	0	10 pts

Question 2

Radical	$5\sqrt{3}$	20 pts

Question 3

x = 7	Left 10; right 17; no	4 pts
x = -7	Left -4; right -11; no	4 pts
x = 0	Left 3; right 3; yes	4 pts
Conclusion	Only the listed matching value works; the equation is not an identity.	8 pts

Question 4

(a)	8i	10 pts
(b)	6	10 pts

Question 5

(a)	$\frac{5}{7}$	10 pts
(b)	$\frac{5}{7}$	10 pts